

HingeEmbeddingLoss#

<https://pytorch.org/docs/stable/generated/torch.nn.HingeEmbeddingLoss.html>

Description:

Measures the loss given an input tensor `xxx` and a labels tensor `yyy` (containing 1 or -1). This is usually used for measuring whether two inputs are similar or dissimilar, e.g. using the L1 pairwise distance as `xxx`, and is typically used for learning nonlinear embeddings or semi-supervised learning. The loss function for `nnn`-th sample in the mini-batch is $\max(0, 1 - \text{margin} \cdot \text{distance})$ and the total loss functions is $\sum_{i=1}^N \max(0, 1 - \text{margin} \cdot \text{distance}_i)$ where $L = \{l_1, \dots, l_N\} \in \mathbb{R}^N$. `margin` (float, optional) – Has a default value of 1.

Function Signature

```
class torch.nn.HingeEmbeddingLoss(margin=1.0,  
size_average=None, reduce=None, reduction='mean')[source]#
```

Parameters

Shape:

```
forward(input, target)[source]#
```

Return type

Returns:

Tensor

Detailed Documentation

Documentation

Measures the loss given an input tensor xxx and a labels tensor yyy (containing 1 or -1). This is usually used for measuring whether two inputs are similar or dissimilar, e.g. using the L1 pairwise distance as xxx, and is typically used for learning nonlinear embeddings or semi-supervised learning.

The loss function for nnn-th sample in the mini-batch is

and the total loss functions is

where $L=\{l_1,\dots,l_N\} \top L = \{l_1,\dots,l_N\}^{\wedge\text{top}L}=\{l_1,\dots,l_N\} \top$.

margin (float, optional) – Has a default value of 1.

size_average (bool, optional) – Deprecated (see reduction). By default, the losses are averaged over each loss element in the batch. Note that for some losses, there are multiple elements per sample. If the field size_average is set to False, the losses are instead summed for each minibatch. Ignored when reduce is False. Default: True

reduce (bool, optional) – Deprecated (see reduction). By default, the losses are averaged or summed over observations for each minibatch depending on size_average. When reduce is False, returns a loss per batch element instead and ignores size_average. Default: True

reduction (str, optional) – Specifies the reduction to apply to the output: 'none' | 'mean' | 'sum'. 'none': no reduction will be applied, 'mean': the sum of the output will be divided by the number of elements in the output, 'sum': the output will be summed. Note: size_average and reduce are in the process of being deprecated, and in the meantime, specifying either of those two args will override reduction. Default: 'mean'

Input: $(*)^{(*)} (*)$ where $*^{**}$ means, any number of dimensions. The sum operation operates over all the elements.

Target: $(*)^{(*)} (*)$, same shape as the input

Output: scalar. If reduction is 'none', then same shape as the input

Runs the forward pass.